

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

14th September 2016

Subject SA2 - Life Insurance

Time allowed: Three hours (14.45* - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. * You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
- 3. You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
- 4. The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
- 5. Attempt all questions, beginning your answer to each question on a separate sheet.*
- 6. Mark allocations are shown in brackets.*
- 7. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1) i)** In the Indian context, explain the differences in the methods of valuation of assets and liabilities between the statutory balance sheet (prepared as per IRDAI requirements) and what might be considered a “realistic balance sheet” of a life insurance company.

You may assume that a realistic balance sheet is defined as the one based on valuation of assets and liabilities on a market-consistent basis.

(15)

An Indian life insurance company sells predominantly non-linked non-participating protection products, comprising mainly term assurances, whole of life policies and immediate annuity products. As part of its internal shareholder reporting, a key performance metric that it reports to its shareholders is the “change in net asset value (measured as total assets minus liabilities)” over the past year, measured on both statutory and “realistic” basis (as defined in Question 1(i) above).

- ii)** Describe the key risk factors that would affect the change in net asset value over a one year period; and how would the impact of each of these factors vary between net asset value measured on the basis of statutory and realistic balance sheets respectively.

(18)

- iii)** The CFO has undertaken an analysis of the reported new business profit margins of the Company, calculated using a Traditional Embedded Value approach. He has concluded that although there are reasonable underwriting and investment margins, the overall profit margins are severely impacted due to high cost of capital. What steps can the Company take to lower its overall cost of capital so that its profit margins improve?

You may assume that the required capital for the Company is currently set equal to the statutory minimum Required Solvency Margin (RSM), as prescribed by the IRDAI.

(11)

- iv)** As part of its diversification strategy, the Company is considering a foray into savings oriented insurance products, as its current product portfolio is significantly weighted towards protection oriented products. The Company is considering designing an individual non-linked non-participating variable insurance product. What are the minimum regulatory requirements in respect of the design and management of such a product that the Company should bear in mind prior to its launch?

(6)
[50]

- Q. 2)** You work for an Indian life insurance company that has been operating for over 10 years. The Company had been known for its success in the pensions business, although following the regulatory changes in the past, the Company lost a bit of momentum in this business. However, after the new product regulations issued by IRDAI in 2013, the Company is looking to revive its prominence in the pensions business.

- i)** State the tax treatment associated with ordinary life annuities from a policyholder’s standpoint. Briefly explain how the tax treatment of ordinary life annuities is favourable or unfavourable vis-à-vis bank deposits.

Note that an ordinary life annuity mentioned above refers to an annuity product that provides a guaranteed income as long as the annuitant is alive but does not return the premium upon the death of the annuitant. Also, there is no guarantee that the annuity will be paid for a certain initial number of years irrespective of whether the annuitant is alive or not.

(7)

- ii) Under the new regulations, policyholders do not have an 'Open Market Option' at the time of utilizing the proceeds from the accumulation pension products to purchase an annuity. In other words, the policyholders must buy annuities from the same insurer which provided the accumulation pension product. As such, providing competitive annuity rates has become more important than before.

The Company offers ordinary life annuities as well as a variant thereof which refunds the purchase price paid, upon death of the annuitant. The Company's distribution team feels that the annuity rates are uncompetitive and pose a challenge in marketing accumulation pension products.

The CEO has asked you to perform an investigation in order to find out ways to improve the competitiveness of the annuity rates. Discuss the various aspects that you will cover in your investigation.

(16)

- iii) During a discussion, the Chief Marketing Officer observed that the variant which refunds the purchase price upon death of the annuitant bears resemblance to fixed interest instruments such as bank deposits. He suggested that the Company should consider offering a guaranteed surrender value equivalent to 95% of the purchase price paid for such a variant.

Discuss the key risks the Company would be exposed to if such a surrender value was offered.

(10)

- iv) The Chief Distribution Officer mentioned that a potential objection from the prospective clients of the pension product is that the annuity rates to be applied to their corpus in the future are not known at the outset. She has suggested that the Company should set forth minimum guaranteed annuity rates, calculated using a 3% interest rate.

She feels that this is a balanced approach, especially considering that the vesting benefit offered under the pensions product already incorporates a 3% p.a. guarantee on regular premiums paid during the accumulation period.

Discuss the merits of her proposal and the risks associated with the same.

(17)

[50]
